

Erdős Problem #304

EP-304

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

For integers $1 \leq a < b$ let $N(a, b)$ denote the minimal k such that there exist integers $1 < n_1 < \dots < n_k$ with $\frac{a}{b} = \frac{1}{n_1} + \dots + \frac{1}{n_k}$. Estimate $N(b) = \max_{1 \leq a < b} N(a, b)$. Is it true that $N(b) \ll \log \log b$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>